PRATIK JAYDEEP KHETMALIS

AI ENGINEER | MACHINE LEARNING ENGINEER | GENERATIVE AI

Pune, India | +91-7387848095 | khetmalisp3@gmail.com | linkedin.com/in/pk6959 | github.com/Pratik0718

PROFESSIONAL SUMMARY

AI/ML Engineer with 2+ years of professional experience in Data Analytics, Python, SQL, and machine learning, with hands-on development of Generative AI and RAG applications. Experienced in end-to-end ML workflows including data preprocessing, feature engineering, model optimization, explainability, experiment tracking, API development, and deployment. Hands-on with LangChain, Hugging Face, embeddings, FAISS, FastAPI, Flask, Streamlit, Scikit-learn, XGBoost, LightGBM, SHAP, and MLflow.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript
Generative AI / RAG: LLMs, RAG, LangChain, Hugging Face, Prompt Templates, Embeddings, FAISS, Sentence Transformers
Machine Learning: Scikit-learn, XGBoost, LightGBM, CatBoost, Logistic Regression, Random Forest, Decision Trees, SVM, KNN
Data & Analytics: Pandas, NumPy, EDA, Data Cleaning, Data Validation, Feature Engineering, SQL Joins, CTEs, Window Functions
Model Optimization: Hyperparameter Tuning, Cross-Validation, ROC-AUC, Precision, Recall, F1-Score, Confusion Matrix, SHAP
APIs / Deployment: FastAPI, Flask, Streamlit, REST APIs, Joblib, Docker
MLOps / Tools: MLflow, Git, GitHub, VS Code, Jupyter Notebook, Google Colab

PROFESSIONAL EXPERIENCE

Tech Mahindra — Associate Software Engineer, Data Analytics | Pune, India | Mar 2024 – Mar 2026

• Developed optimized SQL queries using joins, CTEs, window functions, and aggregations to analyze large business datasets.
• Performed end-to-end exploratory data analysis using Python, Pandas, and NumPy to identify business trends and support data-driven decision-making.
• Automated repetitive reporting processes through Python scripting, reducing manual effort and improving operational efficiency.
• Designed data validation workflows to improve data quality, consistency, and accuracy across analytical pipelines.
• Collaborated with business analysts and cross-functional teams to translate business requirements into technical solutions using Agile methodologies.
• Used Git/GitHub for version control and developed reusable analytical workflows for reporting and data analysis.

Prodigy InfoTech — Machine Learning Intern | Remote | Oct 2023 – Nov 2023

• Built end-to-end data preprocessing pipelines using Python and Scikit-learn.
• Performed data cleaning, feature engineering, feature selection, and model evaluation on real-world datasets.
• Developed classification and regression models and improved performance using hyperparameter tuning and cross-validation.
• Documented the ML model development lifecycle and analytical workflow.

FEATURED PROJECTS

YouTube RAG Chatbot | Python, LangChain, Hugging Face, FAISS, Sentence Transformers, Streamlit

• Built a Retrieval-Augmented Generation chatbot that answers questions from YouTube video transcripts.
• Implemented transcript ingestion, text chunking, embedding generation, vector storage/retrieval with FAISS, and LLM-based response generation.
• Integrated Hugging Face models and Sentence Transformers to build an open-source LLM/embedding workflow without relying on OpenAI APIs.
• Developed an interactive Streamlit interface for conversational querying over video content.

Credit Risk Assessment System | Python, XGBoost, LightGBM, SHAP, MLflow, FastAPI

• Developed an end-to-end credit risk prediction system with automated preprocessing using Scikit-learn ColumnTransformer.
• Engineered financial features including Credit-to-Income and Annuity-to-Income ratios and optimized models using RandomizedSearchCV.
• Applied SHAP for model explainability and MLflow for experiment tracking and model management; exposed predictions through FastAPI.

Customer Churn Prediction System | Python, Scikit-learn, Flask, Joblib

• Built an end-to-end telecom customer churn prediction application covering preprocessing, EDA, feature engineering, model training, and evaluation.
• Compared multiple classification algorithms and selected Gradient Boosting based on model performance; optimized using GridSearchCV.
• Evaluated models using ROC-AUC, Precision, Recall, F1-Score, and Confusion Matrix and deployed predictions through a Flask web application.

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals (AI-900) | Microsoft Certified: Azure AI Engineer Associate (AI-102)

EDUCATION

Bachelor of Engineering (Computer Engineering) — International Institute of Information Technology, Pune | CGPA: 8.2 / 10